

One update step

$$\Gamma(t) = \sum_{k=-K}^K c_k e^{ikt}$$

boundary state, stored band K

$t \in [0, 2\pi)$

$$h(s) = a_0 + \sum_{m=1}^M (a_m \cos ms + b_m \sin ms)$$

scalar normal update, band M

s : arclength angle

$$\delta\Gamma^j = h\nu^j$$

move each point along the normal

ν^j : outward unit normal of Γ^j

$$\Gamma^{j+1} = \Gamma^j + \delta\Gamma^j$$

next accepted state

then stored in band K

In the solver the stored state is the band- K projection of the moved curve after arclength reparametrisation:

$$\Gamma^{j+1} = \Gamma^j + P_K[A(\Gamma^j + h\nu^j) - A(\Gamma^j)] \quad (\text{SC-035 centred state-band update})$$